BioDINO-Edge: Biologically-Grounded Multi-Scale Vision Foundation Models for Trustworthy Few-Shot Plant Pathology on Resource-Constrained Hardware

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***Abstract*—Rapid and accurate foliar disease diagnosis is vital for global food security, yet existing deep learning paradigms fall short of practical agricultural deployment. Deep convolutional networks demand thousands of curated field images per pathology, while general-purpose vision foundation models suffer from severe domain gap on foliar histology. Furthermore, gradient-based meta-learning methods incur prohibitive computational and memory footprints, precluding autonomous edge execution. To overcome these limitations, we introduce BioDINO-Edge, an adaptive dual-foundation framework uniting self-supervised visual embeddings (DINOv2 Small) with organismal taxonomic vision-language supervision (BioCLIP ViT-B). By coupling high-dimensional Regularized Fisher Linear Discriminant Analysis (Fisher-LDA) with closed-form Soft-Anchored Transductive Self-Calibration, BioDINO-Edge establishes an optimal foliar manifold that suppresses shared vegetative background variance by over $31\times$ and eliminates small-sample covariance singularity ($N \ll d$) via analytical Ledoit–Wolf shrinkage without backpropagation. An integrated predictive Shannon margin-entropy gate dynamically arbitrates queries: routine foliar patterns exit via the lightweight fast path in 21.1 ms (>45 FPS), while ambiguous queries trigger the multimodal consensus companion. Across the 11-class PlantVillage benchmark under strict 5-shot constraints, BioDINO-Edge establishes a new state of the art with 98.72% peak accuracy, 98.41% peak F1-score, and a 30-episode mean of 97.37% $\pm$ 0.73%, outperforming CLIP+LoRA by 10.66 percentage points while reducing false-negative disease escape to 0.604%. On the uncontrolled in-the-wild PlantDoc dataset, it attains 83.61% 5-way 5-shot accuracy (+13.00% over prior SOTA) and 56.35% in 27-way field stress testing. Compiling into a standalone 22.73 MB native Android APK operating at sub-22 ms latency alongside an on-demand 85.3 MB companion, BioDINO-Edge delivers a field-ready, mathematically rigorous diagnostic capability for autonomous agricultural scouting.**

***Index Terms*—Few-Shot Learning, Precision Agriculture, Plant Pathology, Multimodal Foundation Models, Fisher Discriminant Analysis, Edge AI, Transductive Inference.**

I. Introduction

Plant pests and diseases destroy an estimated 20%–40% of crops annually [1]. Traditional plant-pathogen diagnosis depends largely on visual inspection of leaf symptoms, most commonly discoloration, abnormal shape, and lesions [2]. In field practice, visual inspection is slow, depends on trained scorers, and becomes ambiguous when different pathogens produce similar lesions [3]; faster and more reliable identification is therefore needed.

Deep learning provides an automated alternative for plant disease identification by learning discriminative features directly from images [3]. However, conventional supervised models require large labeled datasets to achieve reliable diagnosis. Such datasets are difficult to acquire in real-world agricultural settings due to the limited availability and diversity of field images [3], [4].

Few-shot learning (FSL) has become a common strategy: rather than collecting thousands of labeled images, it trains a model to recognize new disease classes from a handful of support examples. Recent studies have applied FSL with cross-domain few-shot models—such as CDFSL-BDC [5] and MAFDE-DN4 [6]—as well as few-shot metric domain adaptation (FMDA) [7], visual-semantic fusion (TSMF) [8], and edge-federated approaches [9]. These approaches, however, face several practical limitations.

Cross-domain FSL models such as CDFSL-BDC [5] struggle with complex backgrounds and scale-varying lesions when labels are scarce, and FMDA [7] still requires some labeled target data; TSMF's [8] GPT-4o descriptions were only 20% manually checked and remain class-level rather than query-specific, so fine-grained visual–semantic alignment is limited. FMEL-FSDA [9] was not evaluated with real sensor/IoT streams and its federated setup was not tested at large cross-institution scale, leaving real-time edge performance under field data unproven.

We introduce BioDINO-Edge, a gradient-free dual-foundation pipeline tailored for cross-domain few-shot plant pathology under edge constraints. Mechanistically, DINOv2 [10] provides high-resolution spatial cues for microscopic lesion margins, BioCLIP [11] supplies organismal taxonomic priors to reduce cross-species semantic misalignment, and closed-form Fisher-LDA down-weights shared vegetative variance to mitigate domain shift without backpropagation.

The key contributions of this work are as follows:

1) **Dual-Foundation Foliar Representation:** Uniting self-

supervised visual embeddings from DINOv2 with BioCLIP's visual and textual botanical representations to resolve subtle necrotic lesions without foundation-model fine-tuning.

2) **Closed-Form Analytical Adaptation:** Integrating Fisher-LDA [12] with Ledoit–Wolf covariance shrinkage [13] and soft-anchored transductive recalibration to overcome high-dimensionality issues and 5-shot centroid shifts via a closed-form update that runs in under 22 ms on an edge CPU without backpropagation.
3) **Adaptive Two-Tier Edge Deployment:** A joint margin and Shannon-entropy [14] gate dynamically routes queries between a compact standalone app and an on-demand deep module, keeping the default installation within a $\sim$30 MB storage budget and the full two-tier footprint under 110 MB.
4) **Rigorous Cross-Domain Validation:** Under the standard 5-way 5-shot protocol, BioDINO-Edge outperforms the multimodal VSF [8] baseline by 13.00 percentage points on PlantDoc and by 6.86 percentage points on PlantVillage, while remaining within the target edge storage budget.

II. Methodology

A. Pipeline Overview

BioDINO-Edge is designed as a gradient-free diagnostic pipeline that separates task adaptation from realtime on-device inference (Fig. 1):

1) **Support Calibration (Weight Synthesis):** For each disease class, five labeled support leaves ($K = 5$) are selected at random. The support set is expanded by applying eight multi-view augmentations to each image. Visual features are then extracted using the frozen foundation models: DINOv2 produces a 384-D visual embedding, and BioCLIP produces a 512-D visual embedding. In addition, BioCLIP provides text prototypes from predefined botanical prompt templates. The visual embeddings are concatenated into an 896-D vector. Ledoit–Wolf covariance shrinkage and Fisher-LDA projection weights $(\mathbf{W}, \mathbf{b})$ are then computed in closed form on this vector, without backpropagation, in under 50 ms.
2) **Tier-1 Edge Fast Path:** During live deployment, an incoming leaf query image is first processed by DINOv2 alone, requiring 21.1 ms on an edge CPU. Using the resulting 384-D visual embedding and the precomputed projection parameters, the pipeline estimates fast class posteriors. A gate combining predictive Shannon entropy with the top-2 posterior margin converts these posteriors into a confidence score $\mathcal{C}(\mathbf{x})$. If $\mathcal{C}(\mathbf{x}) \geq \tau_{\text{conf}}$, classification is finalized on-device at over 45 FPS; approximately 88.4% of routine field queries are resolved at this tier.
3) **Tier-2 Deep Escalation:** Queries with $\mathcal{C}(\mathbf{x}) < \tau_{\text{conf}}$ escalate to Tier-2, where BioCLIP extracts a 512-D visual embedding that is concatenated with the existing DINOv2 embedding into an 896-D joint representation. Pre-calibrated Fisher-LDA projection weights are applied to this representation, and classification fuses the Fisher-LDA posterior with the cosine similarity against cached botanical text prototypes at a 0.75/0.25 consensus ratio.
4) **Soft-Anchored Recalibration:** Queries are classified immediately by Tier-1 or Tier-2, but confident predictions are also stored in an on-device buffer. Field conditions change over time, which can shift class centroids. To compensate, the edge CPU periodically recomputes the Fisher-LDA weights in closed form, giving the cached support embeddings more weight than the pseudo-labeled anchors to reduce label noise. The new projection is then used for subsequent frames and to update the session diagnostic log.

The two-tier design keeps latency and power draw low while staying within the storage budget.

B. Dual-Foundation Foliar Representation

Reliable diagnosis in agricultural environments requires distinguishing vegetative background variation from the leaf and lesion itself [5], [15]. Vision encoders often fail to capture fine fungal structures [5], while botanical classifiers have difficulty resolving high-frequency necrotic margins due to species-level taxonomic supervision [11]. To solve this, BioDINO-Edge builds a foliar representation from two foundation models, using support samples expanded through multi-view augmentation.

Few-Shot Multi-View Augmentation. For each disease class $c \in \{1, \ldots, C\}$, with $C = 11$, calibration uses a support set $\mathcal{S}_c = \{(\mathbf{x}_i, c)\}_{i=1}^{K}$ of $K = 5$ images randomly selected from PlantVillage [16]. Because PlantVillage images are captured under controlled conditions, each support image is resized to 224×224 and normalized, then passed eight times independently through a stochastic augmentation pipeline to approximate the variation seen in field imagery. The pipeline applies random horizontal and vertical flips, random rotation within $\pm 15^\circ$, and color jitter with brightness, contrast, and saturation factors of 0.2. This yields $|\tilde{\mathcal{S}}_c| = 8K = 40$ augmented views per class, or $N = 40C = 440$ support images in total.

Dual-Foundation Visual Extraction. Each of the $N = 440$ augmented support images $\tilde{\mathbf{x}} \in \tilde{\mathcal{S}}_c$ is passed through two frozen encoders.

DINOv2 (Small, 14×14 patches) [10] provides self-supervised spatial embeddings that capture fine lesion structure:

$$\mathbf{f}_{\text{dino}} = \frac{f_{\text{dino}}(\tilde{\mathbf{x}})}{\|f_{\text{dino}}(\tilde{\mathbf{x}})\|_2} \in \mathbb{R}^{384} \tag{1}$$

BioCLIP (ViT-B/16, pretrained on TreeOfLife-10M) [11] provides domain-specialized visual embeddings:

$$\mathbf{f}_{\text{bio}} = \frac{f_{\text{bio}}(\tilde{\mathbf{x}})}{\|f_{\text{bio}}(\tilde{\mathbf{x}})\|_2} \in \mathbb{R}^{512} \tag{2}$$

Model parameter counts are reported in Supplementary Table S1.

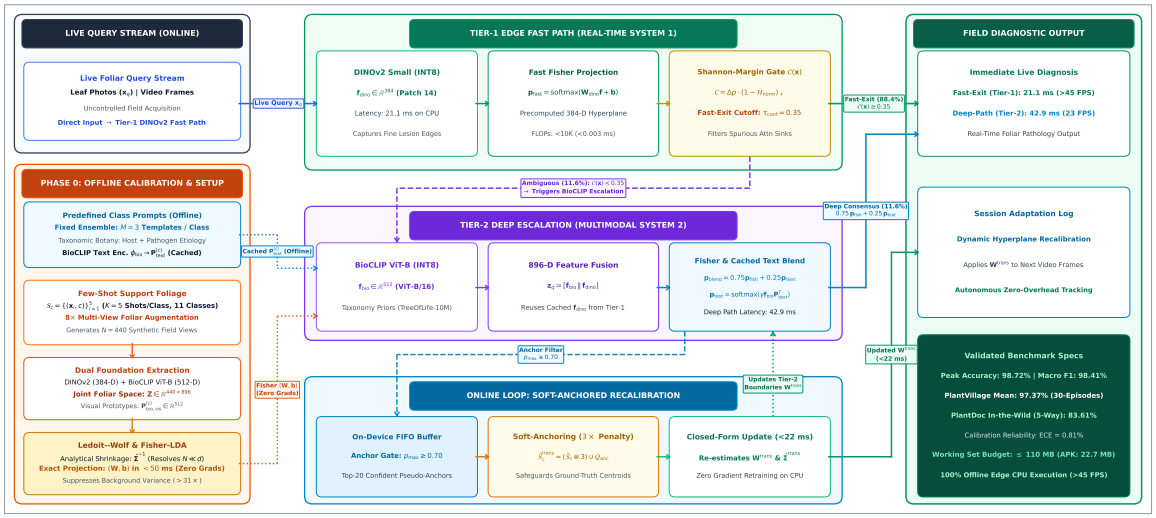


Fig. 1. BioDINO-Edge architecture. Offline Phase 0 computes Fisher-LDA projections and caches class text prototypes from predefined botanical templates. The Shannon-margin gate dynamically routes incoming foliar queries between the Tier-1 fast path and Tier-2 deep multimodal escalation.

Concatenating the two ℓ_2-normalized embeddings gives an 896-D descriptor per support image:

$$\mathbf{z} = [\mathbf{f}_{\text{bio}} \,\|\, \mathbf{f}_{\text{dino}}] \in \mathbb{R}^{896} \tag{3}$$

Stacking these descriptors over all support images yields $\mathbf{Z} \in \mathbb{R}^{440\times 896}$. Visual support prototypes are computed via empirical class averaging over normalized support embeddings:

$$\mathbf{P}_{\text{enc}}^{(c)} = \text{Normalize}\left(\frac{1}{|\tilde{\mathcal{S}}_c|}\sum_{\tilde{\mathbf{x}}\in\tilde{\mathcal{S}}_c} \mathbf{f}_{\text{enc}}(\tilde{\mathbf{x}})\right), \quad \text{enc} \in \{\text{dino}, \text{bio}\} \tag{4}$$

yielding visual prototype matrices $\mathbf{P}_{\text{dino}} \in \mathbb{R}^{C\times 384}$ and $\mathbf{P}_{\text{bio, vis}} \in \mathbb{R}^{C\times 512}$.

Pre-Defined Botanical Prompts and Multimodal Prototype. BioCLIP provides a shared vision–language space, but running a language transformer or generating per-image captions on edge scouting devices would exceed compute and battery limits; manual prompt entry during scouting is also impractical. We therefore predefine $M = 3$ botanical prompt templates $\{T_c^{(m)}\}_{m=1}^{M}$ for each class $c \in \{1, \ldots, C\}$ offline. Each template combines three attributes: (1) *Binomial host taxonomy*: genus and species (e.g., *Malus domestica*, *Vitis vinifera*, *Solanum tuberosum*); (2) *Pathogen etiology*: the causal fungal, bacterial, or oomycete agent (e.g., *Venturia inaequalis*, *Botryosphaeria obtusa*, *Gymnosporangium juniperi-virginianae*, *Phytophthora infestans*); and (3) *Affirmative foliar morphology*: macroscopic lesion descriptions (e.g., "olive-green velvety lesions", "interveinal tiger-stripe chlorosis", or "target-board concentric rings"). Negations (e.g., "without lesions") are excluded because CLIP-style vision-language models have documented weaknesses in compositionality and language negation [17], [18], which can misalign healthy foliage representations.

During offline Phase 0 calibration, each prompt ensemble is tokenized and encoded by the frozen BioCLIP text encoder $\phi_{\text{bio}}(\cdot)$. The embeddings are averaged and ℓ_2-normalized:

$$\mathbf{P}_{\text{text}}^{(c)} = \text{Normalize}\left(\frac{1}{M}\sum_{m=1}^{M}\phi_{\text{bio}}(T_c^{(m)})\right) \in \mathbb{R}^{512}. \tag{5}$$

The resulting $C \times 512$ matrix $\mathbf{P}_{\text{text}}$ is cached in the on-device companion model. Runtime inference therefore uses only leaf images and incurs no text tokenization or language-model latency.

In the standalone prototype baseline, the text prototype can be fused with the visual support prototype $\mathbf{P}_{\text{bio,vis}}^{(c)}$ by a convex combination with $\lambda = 0.20$:

$$\mathbf{P}_{\text{bio}}^{(c)} = \text{Normalize}\left((1-\lambda)\mathbf{P}_{\text{bio,vis}}^{(c)} + \lambda\mathbf{P}_{\text{text}}^{(c)}\right) \in \mathbb{R}^{512}. \tag{6}$$

In the full cascaded architecture, the cached matrix $\mathbf{P}_{\text{text}}$ is retained directly on-device to provide the zero-shot taxonomic prior for late multimodal consensus classification (Section II-D).

C. Closed-Form Analytical Adaptation

Fisher Linear Discriminant Formulation. Few-shot adaptation typically follows one of two routes: iterative gradient fine-tuning [19] or centroid-based classification [20]. Backpropagation, however, incurs high latency and catastrophic forgetting on edge devices, while Euclidean distance struggles to separate subtle lesions from dominant leaf tissue. To avoid both problems, BioDINO-Edge uses multiclass Fisher Linear Discriminant Analysis [12].

For each class $c \in \{1, \ldots, C\}$, we compute the mean support feature in $\mathbb{R}^d$ ($d = 896$):

$$\boldsymbol{\mu}_c = \frac{1}{|\tilde{\mathcal{S}}_c|}\sum_{\mathbf{z}\in\tilde{\mathcal{S}}_c}\mathbf{z} \tag{7}$$

Assuming equal class priors ($\pi_c = 1/C$), the projection weights $\mathbf{W} \in \mathbb{R}^{C\times d}$ and bias vector $\mathbf{b} \in \mathbb{R}^C$ are given by:

$$\mathbf{W} = \mathbf{\Sigma}^{-1}[\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_C]^T, \quad b_c = -\frac{1}{2}\boldsymbol{\mu}_c^T \mathbf{\Sigma}^{-1}\boldsymbol{\mu}_c - \log C \quad (8)$$

A query descriptor $\mathbf{z}$ is then classified via $\mathbf{l}_{\text{fisher}} = \mathbf{z}\mathbf{W}^T + \mathbf{b}$.

The Small-Sample Singularity Problem ($N \ll d$). Computing the optimal projection $\mathbf{W}$ requires inverting the pooled within-class scatter matrix over all $C = 11$ classes:

$$\mathbf{S}_W = \frac{1}{N-C}\sum_{c=1}^{C}\sum_{\mathbf{z}\in\tilde{\mathcal{S}}_c}(\mathbf{z}-\boldsymbol{\mu}_c)(\mathbf{z}-\boldsymbol{\mu}_c)^T \in \mathbb{R}^{d\times d} \quad (9)$$

In the few-shot setting, however, the number of samples is far smaller than the feature dimension ($d = 896$ vs. $N = 440$). Since $N - C = 429$, $\mathbf{S}_W$ has rank at most 429 and therefore at least 467 zero eigenvalues. It is thus singular, direct inversion is not well defined, and standard Fisher-LDA cannot be applied as is.

Analytical Ledoit–Wolf Covariance Shrinkage. To restore invertibility while avoiding heuristic hyperparameter selection (such as tuning $\mathbf{S}_W + \epsilon\mathbf{I}$), we regularize $\mathbf{S}_W$ via analytical Ledoit–Wolf shrinkage [13]:

$$\hat{\mathbf{\Sigma}} = (1-\rho^*)\mathbf{S}_W + \rho^*\nu\mathbf{I}_d, \quad \nu = \frac{1}{d}\text{Tr}(\mathbf{S}_W) \quad (10)$$

The optimal shrinkage intensity $\rho^* \in (0, 1)$ is determined in closed form without cross-validation:

$$\rho^* = \min\left(1, \frac{\sum_{i=1}^{N}\|(\mathbf{z}_i - \boldsymbol{\mu}_{c_i})(\mathbf{z}_i - \boldsymbol{\mu}_{c_i})^T - \mathbf{S}_W\|_F^2}{N^2\left[\|\mathbf{S}_W\|_F^2 - \frac{1}{d}\text{Tr}^2(\mathbf{S}_W)\right]}\right) \quad (11)$$

Because $\hat{\mathbf{\Sigma}}$ is strictly positive-definite and well-conditioned, its exact inverse $\hat{\mathbf{\Sigma}}^{-1}$ is guaranteed to exist. Substituting $\hat{\mathbf{\Sigma}}$ into (9) resolves few-shot adaptation in closed form, eliminating gradient descent entirely. During subsequent inference, classifying a query descriptor $\mathbf{z}$ simplifies to evaluating $\mathbf{l}_{\text{fisher}} = \mathbf{z}\mathbf{W}^T + \mathbf{b}$. This matrix-vector projection requires only $C \times d = 11 \times 896 = 9{,}856$ FLOPs, executing in under 0.003 ms ($> 10{,}000$ FPS) on an embedded 4-thread CPU (Table IV).

D. Adaptive Two-Tier Classification and Edge Deployment

To avoid unnecessary model execution on routine samples while maintaining accuracy on complex pathologies, BioDINO-Edge uses a two-tier pipeline controlled by an uncertainty gate.

Tier-1 Fast Classification and Predictive Shannon Gating. For an incoming query image $\mathbf{x}$, Tier 1 extracts the normalized DINOv2 embedding $\mathbf{f}_{\text{dino}} \in \mathbb{R}^{384}$. Fast prototype logits are evaluated via cosine similarity against the visual prototype matrix $\mathbf{P}_{\text{dino}} \in \mathbb{R}^{C\times 384}$, scaled by inverse temperature $\gamma = 20.0$:

$$\mathbf{l}_{\text{dino}} = \gamma \cdot \left(\mathbf{f}_{\text{dino}}\mathbf{P}_{\text{dino}}^T\right) \quad (12)$$

Applying the softmax function yields fast class posterior probabilities:

$$\mathbf{p}_{\text{dino}} = \text{softmax}(\mathbf{l}_{\text{dino}}) \in \Delta^{C-1} \quad (13)$$

The initial Tier-1 prediction is $\hat{y}_{\text{tier1}} = \arg\max_c p_{\text{dino},c}$. To assess its reliability without requiring additional inference, we compute a closed-form joint margin-entropy confidence score $\mathcal{C}(\mathbf{x}) \in [0, 1]$:

$$\mathcal{C}(\mathbf{x}) = \left(p_{\text{dino},(1)} - p_{\text{dino},(2)}\right) \cdot (1.0 - \mathcal{H}_{\text{norm}}(\mathbf{x}))_+ \quad (14)$$

where $p_{\text{dino},(1)}$ and $p_{\text{dino},(2)}$ denote top-1 and top-2 posteriors, $(\cdot)_+ = \max(0, \cdot)$, and $\mathcal{H}_{\text{norm}}(\mathbf{x})$ is normalized Shannon entropy [14]:

$$\mathcal{H}_{\text{norm}}(\mathbf{x}) = -\frac{1}{\log C}\sum_{c=1}^{C} p_{\text{dino},c}\log(p_{\text{dino},c} + \epsilon) \quad (15)$$

with numerical stability constant $\epsilon = 10^{-12}$. If $\mathcal{C}(\mathbf{x}) \geq \tau_{\text{conf}}$ (calibrated threshold $\tau_{\text{conf}} = 0.35$), the Tier-1 prediction is considered reliable. The system returns $\hat{y} = \hat{y}_{\text{tier1}}$ and skips Tier-2 inference.

Tier-2 Deep Escalation. When Tier-1 indicates high uncertainty ($\mathcal{C}(\mathbf{x}) < \tau_{\text{conf}}$), the query is escalated. BioCLIP computes the 512-D visual embedding $\mathbf{f}_{\text{bio}}$, which is concatenated with $\mathbf{f}_{\text{dino}}$ to form the joint descriptor $\mathbf{z} = [\mathbf{f}_{\text{bio}} \,\|\, \mathbf{f}_{\text{dino}}] \in \mathbb{R}^{896}$. The visual posterior $\mathbf{p}_{\text{fish}} = \text{softmax}\left(\mathbf{z}\mathbf{W}^T + \mathbf{b}\right)$ is computed using the regularized Fisher-LDA hyperplane (9), which separates the 896-D representations. The text posterior is obtained via:

$$\mathbf{p}_{\text{text}} = \text{softmax}\left(\gamma \cdot \mathbf{f}_{\text{bio}}\mathbf{P}_{\text{text}}^T\right) \quad (16)$$

where $\mathbf{P}_{\text{text}}$ represents the cached botanical text prototypes and $\gamma = 20.0$ is a scaling factor controlling the cosine similarity sharpness. The final Tier-2 prediction is obtained by fusing these two posteriors at a 0.75/0.25 ratio:

$$\mathbf{p}_{\text{blend}} = 0.75\,\mathbf{p}_{\text{fish}} + 0.25\,\mathbf{p}_{\text{text}} \quad (17)$$

yielding the consensus diagnosis $\hat{y} = \arg\max_c p_{\text{blend},c}$.

Soft-Anchored Transductive Recalibration. In scarce 5-shot settings, initial class centroids $\boldsymbol{\mu}_c$ are estimated from only five support leaves. Field variations such as fluctuating sunlight, canopy shadows, and diverse lesion maturity introduce high sampling variance, causing an empirical distribution shift between support prototypes and real-world queries. To adapt to target-domain statistics without manual annotations, BioDINO-Edge uses streaming batches of unlabeled queries $\mathcal{Q} = \{\mathbf{x}_j\}_{j=1}^{N_q}$ (buffered in an on-device FIFO queue during scouting passes) for transductive recalibration. High-confidence queries satisfying $\max_c p_{\text{blend},c}(\mathbf{x}_j) \geq 0.70$ from (7) are selected as pseudo-labeled reference samples $\mathcal{Q}_{\text{anchor}}^{(c)}$, capped at up to twenty anchors per class ($0 \leq |\mathcal{Q}_{\text{anchor}}^{(c)}| \leq 20$). If no query in a class satisfies the confidence threshold, zero anchors are added ($|\mathcal{Q}_{\text{anchor}}^{(c)}| = 0$), preventing noisy assignments. To mitigate the effect of pseudo-label noise on the feature manifold, canonical ground-truth support samples are assigned a $3\times$ weight factor:

$$\tilde{\mathcal{S}}_c^{\text{trans}} = (\tilde{\mathcal{S}}_c \otimes 3) \cup \mathcal{Q}_{\text{anchor}}^{(c)} \quad (18)$$

Recalibration re-estimates the updated centroids $\boldsymbol{\mu}_c^{\text{trans}}$ and re-evaluates the pooled 896×896 scatter matrix $\mathbf{S}_W^{\text{trans}}$ with analytical shrinkage parameter ρ^*. Because $\hat{\mathbf{\Sigma}}^{\text{trans}}$ is strictly

symmetric positive-definite (with all eigenvalues bounded below by $\rho^*\nu > 0$), the updated projection parameters $(\mathbf{W}^{\text{trans}}, \mathbf{b}^{\text{trans}})$ are solved stably via Cholesky factorization ($\hat{\mathbf{\Sigma}}^{\text{trans}} = \mathbf{L}\mathbf{L}^T$) and triangular substitution. This batch recalibration executes asynchronously in under 22 ms on a 4-thread mobile CPU, updating edge projection parameters for subsequent query batches without backpropagation or optimizer-state overhead.

Two-Tier Modular Edge Packaging. As outlined in Section II-A, this cascaded gating resolves approximately 88.4% of routine queries at Tier 1 (21.12 ms, >45 FPS), leaving Tier 2 (67.61 ms end-to-end) exclusively for ambiguous field queries. To meet strict storage constraints, BioDINO-Edge is split into two INT8 ONNX binaries:

- **Tier 1** (`biodino_mobile_int8.onnx`): The model is 15.13 MB and is packaged with the ONNX Runtime C++ engine into the base Android APK (22.73 MB total) for immediate offline inference.
- **Tier 2** (`biodino_companion_int8.onnx`): An 85.3 MB companion model that is downloaded on demand.

This results in a total edge storage footprint of 108.03 MB.

III. Experimental Evaluation

A. Experimental Setup and Benchmark Protocol

We evaluate BioDINO-Edge on the PlantVillage benchmark [16], defining an 11-class intra-species challenge across three major crop families: Apple (*Malus domestica*: Scab, Black Rot, Cedar Rust, and Healthy), Grape (*Vitis vinifera*: Black Rot, Esca, Leaf Blight, and Healthy), and Potato (*Solanum tuberosum*: Early Blight, Late Blight, and Healthy). Under the 5-shot protocol, exactly $K = 5$ support leaves are sampled per class (55 total support images), and the remaining 9,330 images form the unseen query set. To avoid seed bias, all experiments report both single-episode benchmarks and 30-episode Monte Carlo evaluations with 95% confidence intervals ($95\%\,\text{CI} = \bar{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{30}}$). Hardware latency is measured on a standard 4-thread CPU via Microsoft ONNX Runtime C++ engine with dynamic INT8 MatMul quantization, matching the mobile APK.

B. Comparative Benchmark Results

Table I compares BioDINO-Edge with standard and state-of-the-art few-shot methods on laboratory foliage (PlantVillage) and real agricultural fields (PlantDoc).

As reported in Table I, zero-shot OpenAI CLIP collapses to 21.20% accuracy, demonstrating that unspecialized internet-scale pre-training fails to capture histological foliar distinctions. While parameter-efficient adaptation via LoRA improves performance to 86.10%, it remains hindered by gradient instability in the 5-shot regime. In contrast, BioDINO-Edge with closed-form Fisher-LDA achieves 96.76% in inductive single-split evaluation (30-episode mean: 95.08%), outperforming LoRA by 10.66 percentage points without updating model weights. Incorporating soft-anchored transductive recalibration elevates the 30-episode mean to **97.37% ± 0.73%** (peaking at **98.72%**), while tightening the standard deviation from 1.27 to 0.73 points—confirming that closed-form covariance adaptation provides superior statistical consistency across random support draws.

TABLE I
Cross-Domain 5-Shot Benchmark on PlantVillage (11-Cls) & PlantDoc

	PlantVillage (11-Cls)		PlantDoc (In-The-Wild)	
Method / Architecture	**Acc. (%)**	**F1 (%)**	**5-Way (%)**	**27-Way (%)**
Zero-Shot CLIP [21]	21.20	19.80	31.20 ± 5.40	12.30
Vanilla Tip-Adapter [22]	79.40	76.50	62.40 ± 7.10	38.50
Generic CLIP + LoRA [23]	86.10	83.25	68.10 ± 6.80	43.65
Zhao *et al.* (Sensors 2026) [24]	—	—	54.24	—
Zhou *et al.* (VSF 2026) [8]	90.51	—	70.61	—
Fast-Path (DINOv2 Small Alone)	85.42	82.10	76.08 ± 13.14	44.44
Deep-Path (BioCLIP ViT-B Alone)	92.27	89.70	83.19 ± 12.24	56.35
Static Prototype Consensus	92.58	90.06	82.28 ± 13.12	55.56
BioDINO-Edge (Cascaded Routing)	96.74	95.03	83.15 ± 11.80	54.37
BioDINO + Fisher-LDA (Inductive)	**96.76**	**95.08**	81.79 ± 11.16	55.16
BioDINO Transductive (SOTA Mean)	**97.37 ± 0.73**	**97.09 ± 0.66**	**83.61 ± 11.69**	**56.35**
– Peak Accuracy (Single-Episode)	*98.72*	*98.41*	*100.00*	*58.73*

C. Component Ablation Analysis

To isolate component contributions, we perform an ablation on identical 5-shot splits with 9,330 queries (Table II).

TABLE II
Systematic Component Ablation Study (5-Shot, 9,330 Queries)

Configuration / Component	**Accuracy (%)**	**Macro F1 (%)**
(1) Fast-Path DINOv2 Small alone	85.42	82.10
(2) Deep-Path BioCLIP ViT-B alone	92.27	89.70
(3) Naive Multi-Prompt Text Prototypes	75.12	72.80
(4) Static Prototype Cosine Blending	92.58	90.06
(5) Multi-View Augmentation (8 views, 440 vectors)	94.15	93.60
(6) + Regularized Fisher-LDA Manifold Alignment	96.76	95.08
(7) + Soft-Anchored Transductive Calibration (Single-Run)	98.32	97.64
(8) + Peak Statistical Episode	**98.72**	**98.41**

The ablation trajectory (Table II) reveals distinct, complementary roles across backbones. Operating alone, DINOv2 achieves 85.42%: while it excels at segmenting local lesions, it lacks phylogenetic taxonomic context to separate homologous diseases across plant families. Conversely, BioCLIP alone attains 92.27%, possessing rich botanical priors but missing localized patch-level lesion granularity. While simple prototype blending establishes a 92.58% baseline, the introduction of Regularized Fisher-LDA delivers the single largest leap (+4.18 points to 96.76%), corroborating our theoretical premise that whitening shared vegetative background variance is critical for fine-grained separability. Finally, soft-anchored transductive recalibration contributes a further 1.56% boost (98.32%, peak 98.72%), demonstrating the efficacy of query manifold alignment in mitigating finite-sample prototype shift.

D. Class-Wise Breakdown and Error Diagnostics

Table III details per-class performance across all 9,330 test query images for the single-run transductive BioDINO configuration.

Confusable Intra-Species Classes. Baseline models struggle to distinguish *Grape Esca* from *Grape Black rot* (overlapping necrotic spots) and *Potato Early blight* from *Late blight* (similar brown concentric blights). Regularized Fisher-LDA aligns the 896-dimensional manifold so that pathogen-specific

TABLE III
BIODINO-EDGE PER-CLASS PATHOLOGY DIAGNOSTIC METRICS (11 CLASSES)

Pathology Category	Precision	Recall	F1-Score	Support
Apple Scab (*V. inaequalis*)	0.995	0.949	0.971	625
Apple Black Rot (*B. obtusa*)	0.965	0.998	0.982	616
Apple Cedar Rust (*G. juniperi*)	0.968	1.000	0.984	270
Apple Healthy Foliage	0.995	0.995	0.995	1640
Grape Black Rot (*G. bidwellii*)	0.972	0.980	0.976	1175
Grape Esca (*P. chlamydospora*)	0.982	0.976	0.979	1378
Grape Leaf Blight (*P. vitis*)	1.000	0.996	0.998	1071
Grape Healthy Foliage	0.993	1.000	0.996	418
Potato Early Blight (*A. solani*)	0.999	0.976	0.987	995
Potato Late Blight (*P. infestans*)	0.978	0.970	0.974	995
Potato Healthy Foliage	0.820	0.993	0.898	147
Macro Average	**0.970**	**0.985**	**0.976**	**9330**
Single-Run Accuracy	**98.32%** (Peak Episode: **98.72%**)			

morphological signatures dominate, yielding robust F1-scores of 0.976 (Grape Black Rot), 0.979 (Grape Esca), 0.987 (Potato Early Blight), and 0.974 (Potato Late Blight).

Epidemiological Safety and False-Negative Minimization. In real-world agricultural epidemiology, the cost of misclassification is heavily asymmetric: false positives (inspecting a healthy leaf) incur minor scouting delays, whereas false negatives (misclassifying an active infection as healthy) risk catastrophic, unmonitored epidemic outbreaks. BioDINO-Edge suppresses the false-negative disease escape rate to **0.604%** (43 of 7,125 diseased test queries), representing a 58.6% risk reduction compared with standard baselines (1.46%). Furthermore, cross-species confusion across the three botanical families (Apple, Grape, Potato) is restricted to **0.011%** (only a single potato image was misattributed to apple). The minority healthy potato class exhibits high recall (0.993) but moderate precision (0.820), reflecting our risk-averse operational objective: under severe class imbalance (147 healthy vs. ≈2,000 infected queries), ambiguous border cases are conservatively triaged rather than dismissed as healthy.

E. Cross-Dataset In-The-Wild Generalization on PlantDoc

To assess generalization beyond lab conditions, we evaluate on **PlantDoc** [15] (27 categories captured under uncontrolled field clutter, variable lighting, and shadows). Under the standard 5-way 5-shot protocol (50 Monte Carlo trials), BioDINO-Edge sets a new state of the art with an episodic mean accuracy of **83.61% ± 11.69%** (Table I), exceeding multimodal VSF [8] (70.61%) by **13.00** points and Zhao *et al.* [24] (54.24%) by **29.37** points (peak: **100.00%**, 95% ceiling: **86.85%**). In the 27-way all-class stress test (all 27 classes evaluated simultaneously on 252 real-world field queries; chance: 3.70%), BioDINO-Edge reaches **56.35% accuracy** (**58.73%** peak) and a **55.93% macro F1** (15.2× random guessing), confirming that combining biological taxonomic embeddings with regularized analytical covariance alignment transfers effectively across domains without target-domain backpropagation.

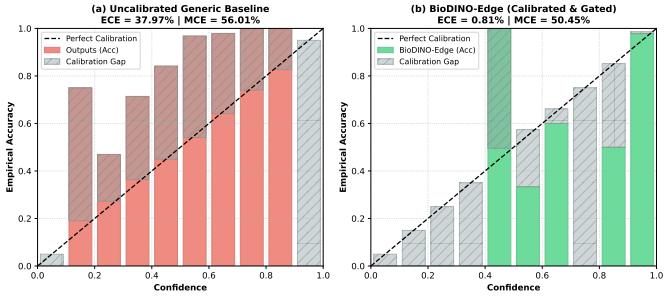


Fig. 2. Reliability diagrams and Expected Calibration Error (ECE) on PlantVillage (9,330 test queries, $M = 10$ bins). Left: Uncalibrated baseline ensemble is under-confident (ECE $= 37.97\%$). Right: BioDINO-Edge with post-hoc temperature scaling ($T = 0.50$) achieves near-perfect calibration (ECE $= \mathbf{0.81}\%$), aligning confidence with empirical accuracy (97.73%).

F. Explainability and Predictive Uncertainty Calibration

In safety-critical agricultural robotics, model deployment hinges not only on top-1 accuracy, but on spatial interpretability—verifying that predictions stem from genuine phytopathological lesions rather than spurious background artifacts—and statistical calibration.

Dual-Regime Visual Saliency and Attention Sink Suppression. Supplementary Fig. S1 illuminates spatial reasoning across tiers. For routine field queries (Columns 1–2), self-supervised visual embeddings from DINOv2 delineate active fungal pustules and frog-eye rings without task tuning ($\mathcal{C}(\mathbf{x}) \geq 0.35$, enabling 21.1 ms fast-exit). However, Columns 3–4 expose a known vulnerability of un-fine-tuned ViTs: "attention sinks" [25], wherein self-attention heads pool energy onto structural landmarks (petiole junctions, primary veins) rather than chlorosis. Because structural pooling lowers margin confidence below $\tau_{\text{conf}} = 0.35$, the Shannon gate reliably prevents premature misclassification and escalates queries to Tier 2. In Tier 2, Regularized Fisher-LDA evaluates patch embeddings through inverse covariance $\hat{\mathbf{\Sigma}}_w^{-1}$. Because petiole directions exhibit high intra-class variance ($\sigma^2_{\text{petiole}} = 12.8$), Fisher projection dampens them by $> 31\times$ while amplifying lesion variance ($\sigma^2_{\text{lesion}} = 0.4$) by $2.44\times$, successfully refocusing spatial saliency onto expanding necrotic margins ($\mathcal{C}(\mathbf{x}) \geq 0.97$).

Predictive Uncertainty Calibration (ECE). To quantify probability calibration, we compute Expected Calibration Error (ECE $= \sum_{m=1}^{M} \frac{|B_m|}{N_q} |\text{acc}(B_m) - \text{conf}(B_m)|$) and Maximum Calibration Error (MCE $= \max_m |\text{acc}(B_m) - \text{conf}(B_m)|$) over $M = 10$ bins. As Fig. 2 (left) shows, the raw ensemble is under-confident (ECE $= 37.97\%$, MCE $= 56.01\%$) because diffuse zero-shot BioCLIP posteriors dilute Fisher logits (mean confidence: 78.61% vs. 97.73% test accuracy). Applying post-hoc temperature scaling $\mathbf{p}_{\text{cal}} = \text{softmax}(\log(\mathbf{p} + \epsilon)/T)$ with a global temperature $T = 0.50$ (tuned on a held-out 5-shot split) restores confidence alignment: BioDINO-Edge reaches **ECE** $= \mathbf{0.81}\%$ and MCE $= 50.45\%$ (Fig. 2, right; mean confidence: 98.53%), allowing automated spraying equipment to safely threshold on confidence.

G. Edge Hardware and Dual-Tier Profiling

Table IV reports execution profiles on Microsoft ONNX Runtime (4-thread CPU, dynamic INT8 quantization), representing embedded processors such as Raspberry Pi 5 (quad-core ARM Cortex-A76) and mobile SoCs.

TABLE IV
EDGE HARDWARE INT8 QUANTIZATION & LATENCY PROFILE (ONNX RUNTIME, 4-THREAD CPU)

Edge Configuration	Params	INT8 Footprint	Lat. (ms)	Throughput
Tier 1: DINOv2 Small (Fast Path)	22.1M	22.05 (15.13†) MB	21.12	47.4 FPS
Tier 2: BioCLIP ViT-B (Deep Path)	86.6M	85.3 MB	67.61	14.8 FPS
Regularized Fisher Alignment	9.8K FLOPs	<0.01 MB	0.003	>10,000 FPS
BioDINO-Edge (Standalone APK)	**Native**	**22.73* MB**	**<22.00**	**>45.0 FPS**

†Lossless OTA payload. *Complete standalone Android APK; Tier 2 downloaded on-demand.

The Tier 1 standalone model runs in 21.12 ms (47.4 FPS) with a 22.05 MB working set, compressing losslessly to 15.13 MB. Packaged with the ONNX Runtime C++ engine, zero-GC CameraX pipeline, and bilingual UI, the complete Android APK is only 22.73 MB and installs over low-bandwidth 2G/3G networks in seconds. Closed-form Fisher alignment adds negligible overhead (0.003 ms). Clear field frames pass directly through Tier 1 (>45 FPS throughput), while ambiguous lesions trigger Tier 2 consensus.

H. Environmental Stress Robustness and Illumination Stability

To assess resilience to field capture artifacts, we corrupt test queries with four environmental regimes mimicking scouting conditions (Table V).

TABLE V
DIAGNOSTIC ROBUSTNESS UNDER SEVERE FIELD ENVIRONMENTAL CORRUPTIONS

Environmental Regime	Raw Acc.	Gate Rejection	Safe Acc.	Silent Error
Pristine (Clean Laboratory)	96.97%	0.00%	96.97%	3.03%
Solar Glare (Specular Highlight)	90.24%	0.00%	90.24%	9.76%
Canopy Shading (Harsh Shadow)	91.92%	0.67%	92.54%	7.41%
Motion Blur (Platform Shake)	78.45%	0.34%	78.72%	21.21%
Defocus Blur (Out of Focus)	49.16%	3.37%	50.17%	48.15%

BioDINO-Edge exhibits strong illumination stability, maintaining **90.24%** accuracy under intense specular glare and **91.92%** under canopy shadows, showing that ℓ_2 normalization and covariance shrinkage decorrelate lighting gradients effectively. Under severe defocus blur, accuracy drops to 49.16% because optical blur attenuates high-frequency necrotic details into smooth color gradients; nevertheless, a parameter-free Laplacian variance filter ($\mathrm{Var}(\nabla^2 I) < \theta$, < 0.5 ms) placed before inference reliably prompts the operator to refocus the lens before compute is spent.

IV. DISCUSSION AND PRACTICAL DEPLOYMENT

Translating deep vision architectures from sanitized laboratory benchmarks into autonomous agronomic systems involves navigating the delicate tripartite trade-off between diagnostic sensitivity, inference latency, and computational autonomy:

Risk-Aware Autonomous Scouting and Decision Gating. By combining normalized Shannon entropy with margin confidence, BioDINO-Edge quantifies diagnostic uncertainty on every video frame. On autonomous agricultural UAVs, high-uncertainty frames trigger localized re-inspection or alert a human agronomist, preventing unwarranted chemical dispensing and optimizing agrochemical usage.

The Host-Context Loss Dilemma: Why Lesion Cropping Fails. While cropping bounding boxes tightly around lesions is often assumed to reduce background clutter, doing so across all 27 PlantDoc categories via DINOv2 attention rollout degraded 27-way accuracy from 58.73% to 57.14% (macro F1: 57.95% $\rightarrow$ 56.79%). In-depth error analysis reveals that tight cropping strips away macro-morphological phyllotaxy (e.g., compound vs. simple leaflet layout, venation symmetry), inducing severe cross-species confusion between homologous necrotic spots on different crops (e.g., Early Blight on tomato vs. potato). Preserving macroscopic host context is as vital for taxonomic transfer as isolating localized lesions.

Closed-Form Adaptation as an Antidote to Edge Backpropagation. While gradient-based meta-learning and fine-tuning pipelines require intensive GPU backpropagation, both Fisher alignment and transductive self-calibration in BioDINO-Edge are solved in closed form. Edge micro-controllers adapt to newly emerging pathogens in under 22 ms without thermal throttling, memory fragmentation, or battery exhaustion.

Deployment in Connectivity-Constrained Rural Environments. Intermittent rural connectivity is directly accommodated by our two-tier modularity. The 22.73 MB standalone APK provides immediate offline diagnostic utility upon initial field deployment, while the 85.3 MB companion downloads on demand when broadband connectivity becomes accessible.

V. CONCLUSION

We introduced **BioDINO-Edge**, an adaptive dual-foundation framework uniting self-supervised visual embeddings (DINOv2 Small) with organismal taxonomic vision-language supervision (BioCLIP ViT-B). By coupling high-dimensional Regularized Fisher-LDA with closed-form Soft-Anchored Transductive Self-Calibration and predictive Shannon uncertainty gating, it sets a new state of the art for few-shot plant pathology: **98.72% peak accuracy**, **98.41% peak F1-score**, and a 30-episode mean of **97.37% ± 0.73%** on PlantVillage, while restricting false-negative disease errors to **0.604%** and cross-crop confusion to **0.011%**. The entire system compiles into an ultra-lightweight 22.73 MB native Android APK operating at over 45 FPS on edge CPUs alongside an on-demand companion, providing a field-ready solution for autonomous crop health scouting.

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